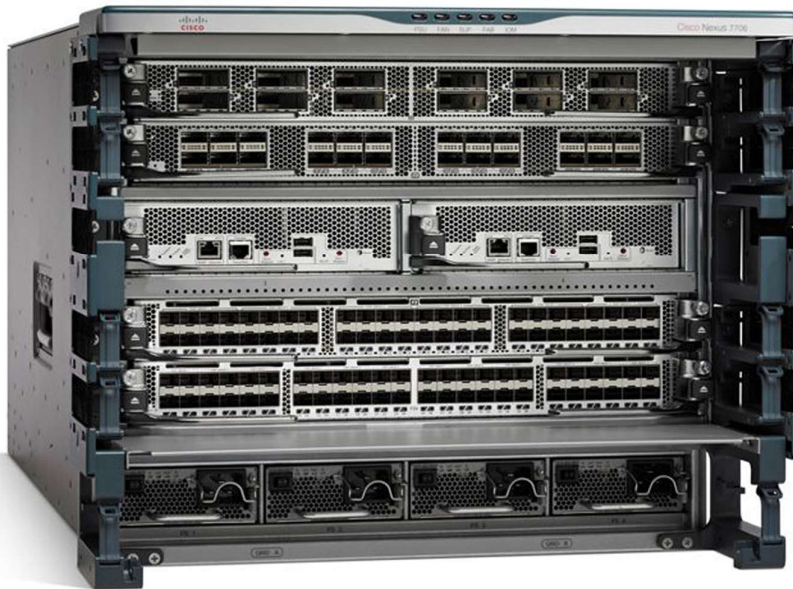




Welcome to CCNA

> 200-301



M.Reza “Ebrahimi”

CCIE# 65249

✉ Ebrahimi@zoombyte.edu.af

in [Linkedin.com/in/mrt1178/](https://www.linkedin.com/in/mrt1178/)



CCNA class in Zoom Byte

- CCNA 200-301
- Full outline
- Video training (4k & Full HD)
- Lab Practice
- Practically training
- Included all software's
- Training support
- Official reference Book
- Passing online guide
- 100% Passing Guarantee

Copyright © ZoomByte | CCNA



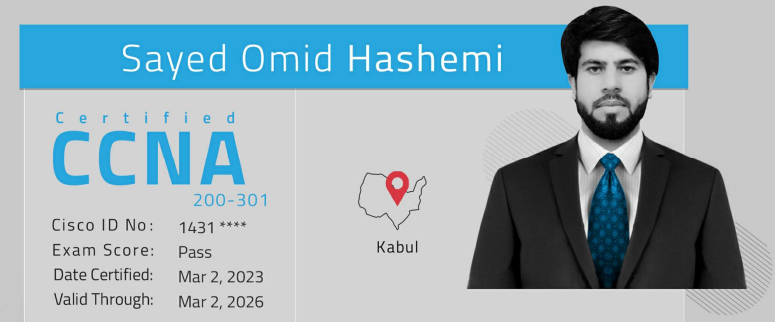
Pearson VUE
Authorised Test Centre

Networking
cisco Academy

CCNA Exam Information

- Exam Name: Cisco CCNA with 200-301 exam code
- Only one exam with no Pre-requisites
- Number of Questions: 100-110
- Exam time: 120-150 Minutes
- Exam Fee: 300\$
- English or Japanese language
- Validity duration: 3 Years
- 825/1000 score at least
- Simulation Questions
- Single choice, Multi-choice , Drag & Drop
- Scenario and exhibit likes simulations
- No calculator and previsions button
- Result at the End

Copyright © ZoomByte | CCNA



Our Recommendations

- **3x** times practice each lecture
- Complete CCNA 200-301 Class
- Complete & Practice All Technologies
- Study the CCNA Cert. Guide for more
- Focus on knowledge gaps
- Study & Practice Passing online toolkit
- Set a date and be committed
- Don't stop at CCNA, go ahead
- Think positives!



Cisco Certification Path



Associate

Master the essentials needed to launch a rewarding career and expand your job possibilities with the latest technologies.

Professional

Select a core technology track and a focused concentration exam to customize your professional-level certification.

Expert

This certification is accepted worldwide as the most prestigious certification in the technology industry.

Certification Requirements:

[CCNA](#)

200-301 CCNA

Certification Requirements:

[CCNP Enterprise](#)

Core exam:
350-401 ENCOR

(Choose one) Concentration exam:

300-410 ENARSI
300-415 ENSDWI
300-420 ENSLD
300-425 ENWLSL
300-430 ENWLSI
300-435 ENAUTO

[CCNP Service Provider](#)

Core exam:
350-501 SPCOR

(Choose one) Concentration exam:

300-510 SPRI
300-515 SPVI
300-535 SPAUTO

[CCNP Data Center](#)

Core exam:
350-601 DCCOR

(Choose one) Concentration exam:

300-610 DCID
300-615 DCIT
300-620 DCACI
300-625 DCSAN
300-630 DCACIA
300-635 DCAUTO

[CCNP Security](#)

Core exam:
350-701 SCOR

(Choose one) Concentration exam:

300-710 SNCF
300-715 SISE
300-720 SESA
300-725 SWSA
300-730 SVPN
300-735 SAUTO

[CCNP Collaboration](#)

Core exam:
350-801 CLCOR

(Choose one) Concentration exam:

300-810 CLICA
300-815 CLACCM
300-820 CLCEI
300-820 CLCNF
300-835 CLAUTO

Certification Requirements:

[CCIE Enterprise Infrastructure](#)

350-401 ENCOR
CCIE Enterprise Infrastructure v1.0 lab

[CCIE Service Provider](#)

350-501 SPCOR
CCIE Service Provider v5.0 lab

[CCIE Security](#)

350-701 SCOR
CCIE Security v6.0 lab

[CCDE](#)

400-007 CCDE Written Exam
CCDE v3.0 Practical Exam

[CCIE Enterprise Wireless](#)

350-401 ENCOR
CCIE Enterprise Wireless v1.0 lab

[CCIE Data Center](#)

350-601 DCCOR
CCIE Data Center v3.0 lab

[CCIE Collaboration](#)

350-801 CLCOR
CCIE Collaboration v3.0 lab

Certification Requirements:

[Cisco Certified DevNet Associate](#)

200-901 DEVASC

Certification Requirements:

[Cisco Certified DevNet Professional](#)

Core exam:
350-901 DEVCOR

(Choose one) Concentration exam:

300-910 DEVOPS
300-915 DEVIOT
300-920 DEVWBX
300-435 ENAUTO
300-535 SPAUTO
300-635 DCAUTO

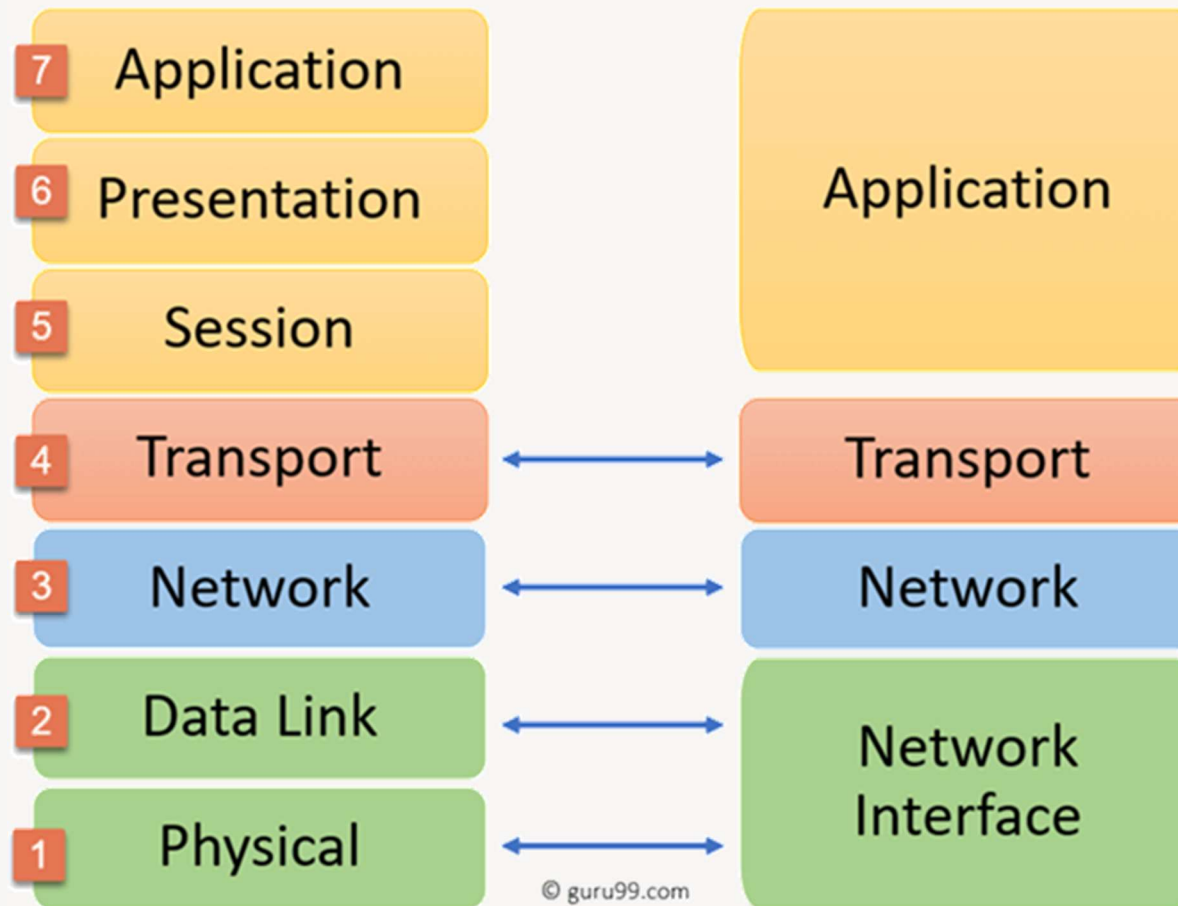
300-735 SAUTO
300-835 CLAUTO

Certification Requirements:

[Cisco Certified DevNet Expert](#)

350-901 DEVCOR
DevNet Expert v1.0 lab

OSI and TCP/IP Models



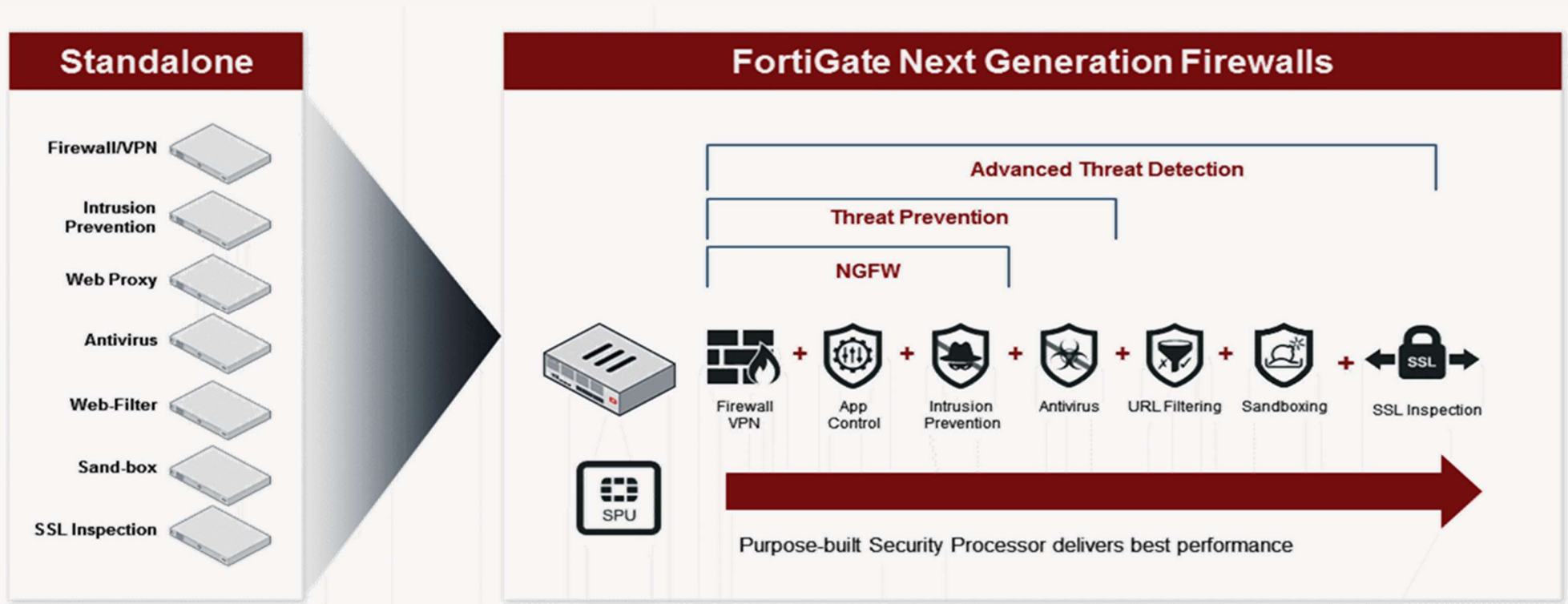


Introduction to Networking Devices

- Switch Layer 2 & Layer 3
- Routers: **Expandability: Media: Operating system features:**
- Access Points & Wireless Routers
- Wireless LAN Controller
- Cisco VOIP
- Firewalls: Software or Hardware | stateful connections
- Servers
- Medias
- Network Printers
- Security Cameras
- NVRs
- Cisco Telepresence



Next Generation Firewall





Next Generation Firewall

Traditional firewall: An NGFW performs traditional firewall functions, such as stateful firewall filtering, NAT/PAT, and VPN termination.

Application Visibility and Control (AVC): AVC makes it possible to look deeply into the application layer data to identify the application to defend against attacks that use random port numbers.

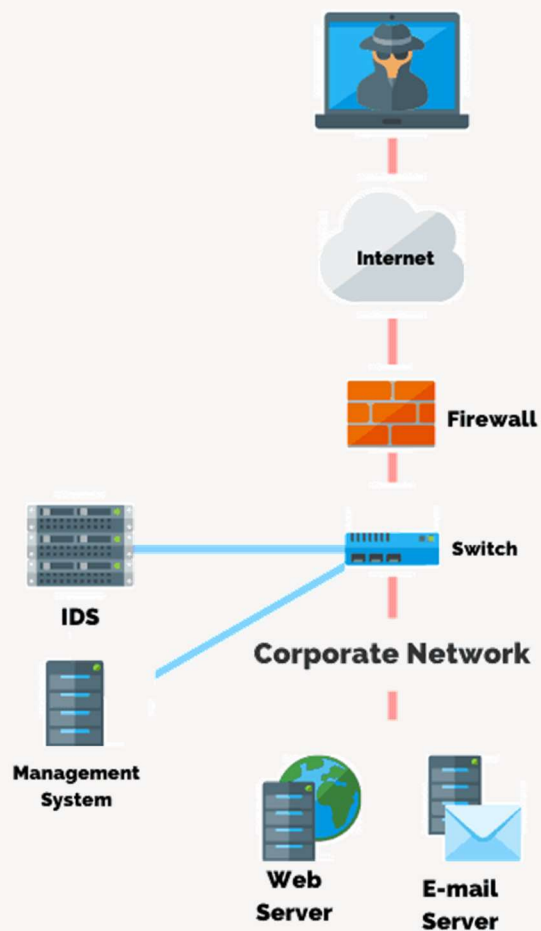
Advanced Malware Protection (AMP): AMP can block file transfers that would install malware and save copies of files for later analysis.

Uniform resource locator (URL) filtering: URL filtering examines the URLs in each web request, categorizes the URLs, and either filters or rate limits the traffic based on rules. The Cisco Talos security group monitors and creates reputation scores for each domain known in the Internet, and URL filtering can use those scores in its decisions to categorize, filter, or rate limit.

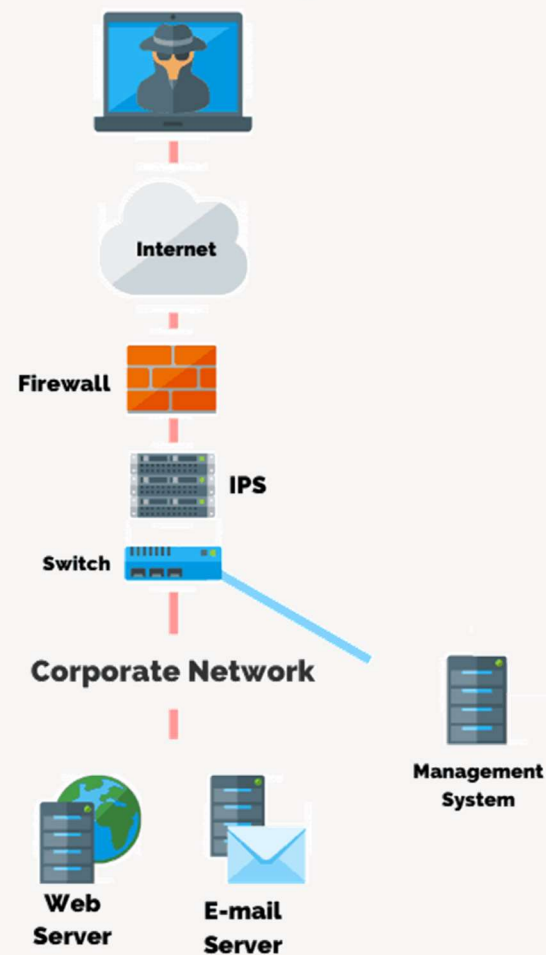
NGIPS: Cisco's NGFW products can also run their NGIPS feature along with the firewall.

IPS vs IDS

Intrusion Detection System (IDS)



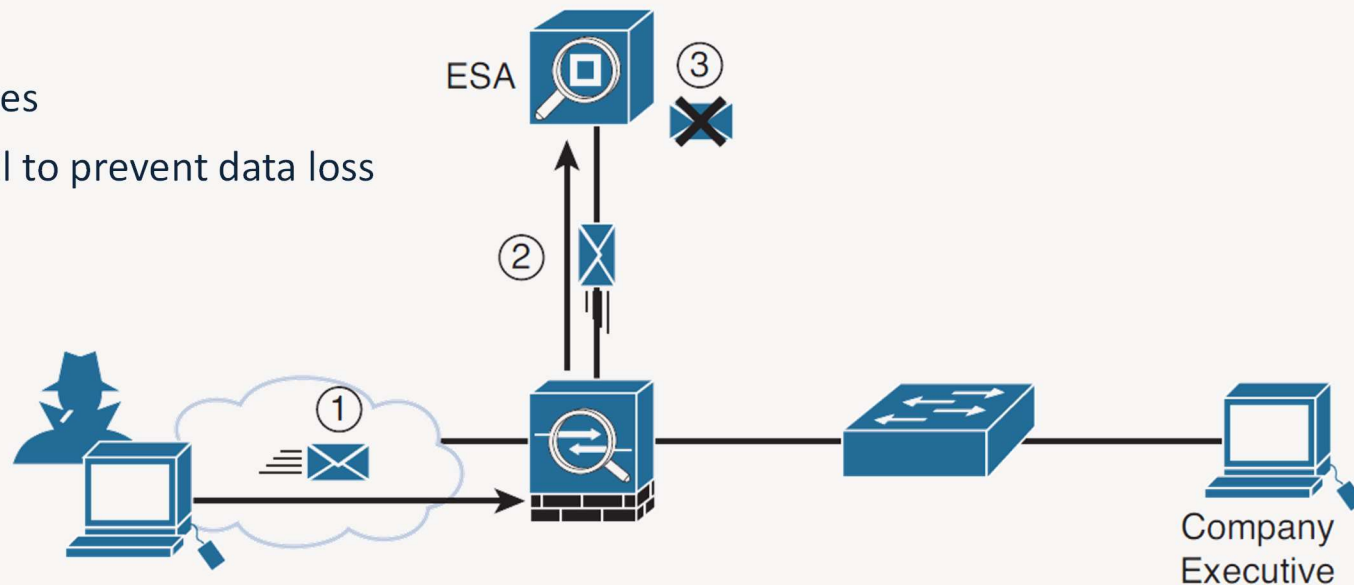
Intrusion Prevention System (IPS)



VS

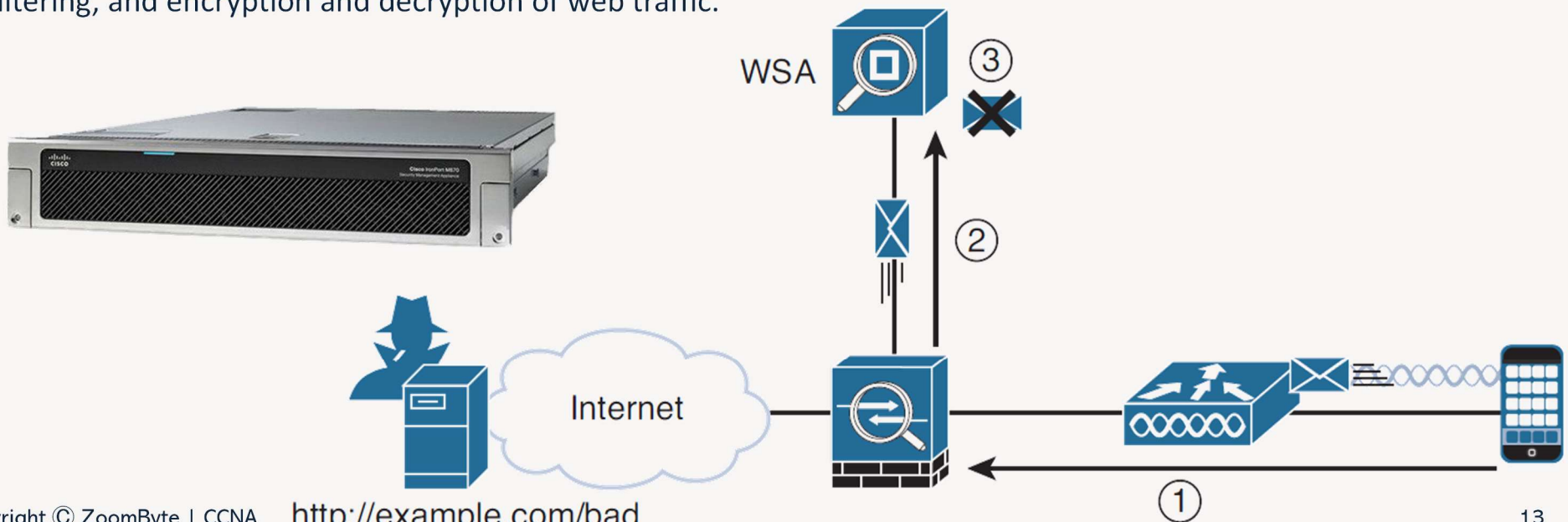
Email Security Appliance (EMA)

- Cisco ESA is special device designed to monitor email's primary protocol, (SMTP). Cisco ESA can do the following:
- Block known threats
- Against stealth مخفيانه malware
- Discard emails with bad links
- Block access to newly infected sites
- Encrypt content in outgoing email to prevent data loss



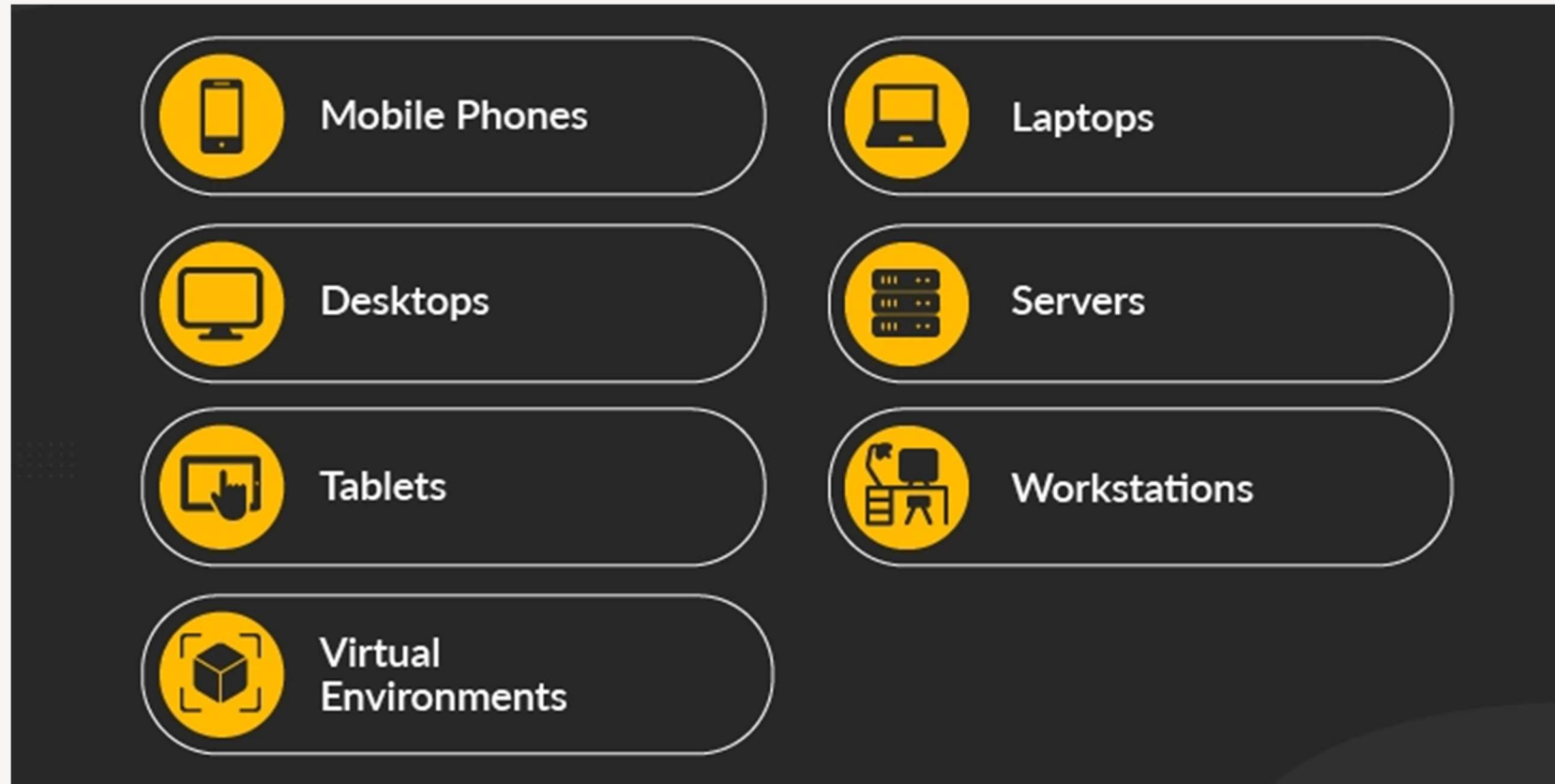
Web Security Appliance (WSA)

- Cisco WSA combines advanced malware protection, application visibility and control, acceptable use policy controls, and reporting. Cisco WSA provides complete control over how users access the Internet. Certain features and applications, such as chat, messaging, video, and audio can be allowed, restricted with time and bandwidth limits, or blocked, according to the organization's requirements.
- WSA can perform blacklisting of URLs, URL filtering, malware scanning, URL categorization, Web application filtering, and encryption and decryption of web traffic.





Endpoint Devices





Servers





Media Types

- Coaxial
- Twisted pair
- Fiber optic
- Wireless
- Cross & straight cable
- LAN & WAN cable
- Cable length:
- Cost:
- Bandwidth:
- Ease of installation:
- Susceptible to EMI/RFI:

Ethernet Type	Bandwidth	Cable Type	Maximum Distance
10BASE-T	10 Mbps	Cat3/Cat5 UTP	100 m
100BASE-TX	100 Mbps	Cat5 UTP	100 m
100BASE-TX	200 Mbps	Cat5 UTP	100 m
100BASE-FX	100 Mbps	Multimode fiber	400 m
100BASE-FX	200 Mbps	Multimode fiber	2 km
1000BASE-T	1 Gbps	Cat5e UTP	100 m
1000BASE-TX	1 Gbps	Cat6 UTP	100 m
1000BASE-SX	1 Gbps	Multimode fiber	550 m
1000BASE-LX	1 Gbps	Single-mode fiber	2 km
10GBASE-T	10 Gbps	Cat6a/Cat7 UTP	100 m
10GBASE-SX4	10 Gbps	Multimode fiber	550 m
10GBASE-LX4	10 Gbps	Single-mode fiber	2 km

Network Device Icon



Printer



PC



Laptop



Server



IP Phone



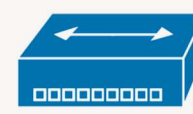
Router



Switch



Layer 3 Switch



Hub



Bridge



Access Point



ASA



Network Cloud



Cable Modem



CSU/DSU



Cable (Various)



Serial Line



Virtual Circuit



Ethernet WAN

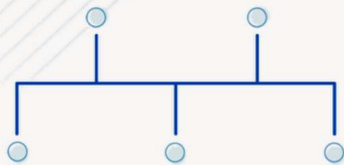


Wireless

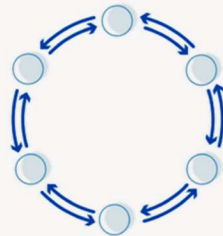


Network Topology

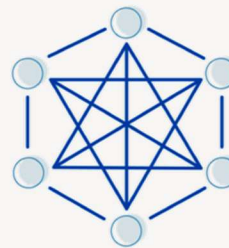
Bus Topology



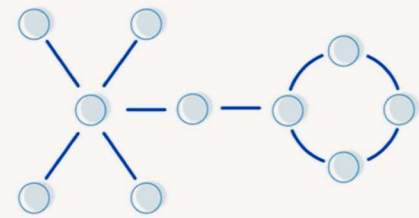
Dual Ring Topology



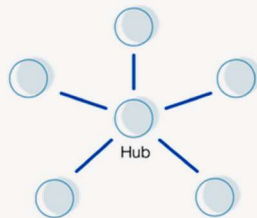
Mesh Topology



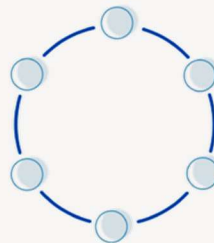
Hybrid Topology



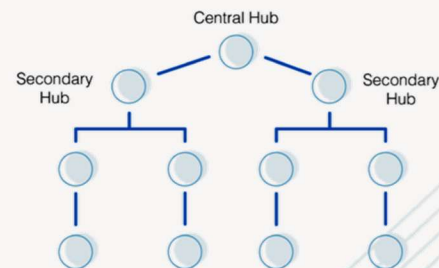
Star Topology



Ring Topology



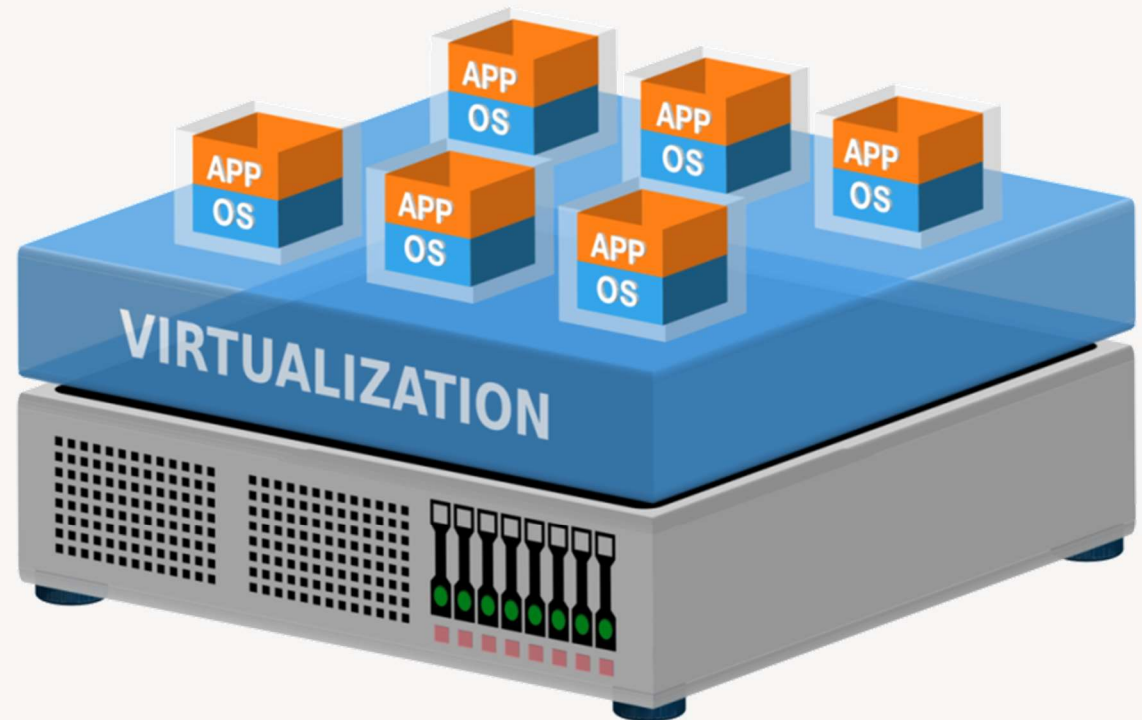
Tree Topology





Virtualization

- Virtualization benefits
- Host OS | Guest OS
- Snapshot or Checkpoint
- Cluster
- Type 1 and Type 2 Hypervisor
- Nested Virtualization
- Container and Docker virtualization
- Virtual Switches and Virtual NICs
- AWS, Microsoft, AliBaba ... VMs





Introduction to Packet Tracer





Basic Configuration

- Help commands
- Command modes
- Verifying commands
- Resetting switch configuration
- Setting host names
- Setting passwords
- Setting IP addresses and default gateways
- Setting interface descriptions
- The **mdix auto** command
- Setting duplex operation
- Setting operation speed
- Managing the MAC address table
- Configuration example
- Shortcuts for entering commands
- Using the **Tab** key to enter complete commands
- Console error messages
- Using the question mark for help
- **enable** command
- **exit** command
- **end** command
- **disable** command
- **logout** command
- Setup mode
- Keyboard help
- History commands
- **terminal** commands
- **show** commands
- Using the pipe parameter (|) with the **show** or **more** commands
- Using the **no** and **default** forms of commands



Power Over Ethernet

- Negotiation the Power Inline with devices
- No need for power adapter
- Central Backup (UPS)
- CDP & LLDP negotiate

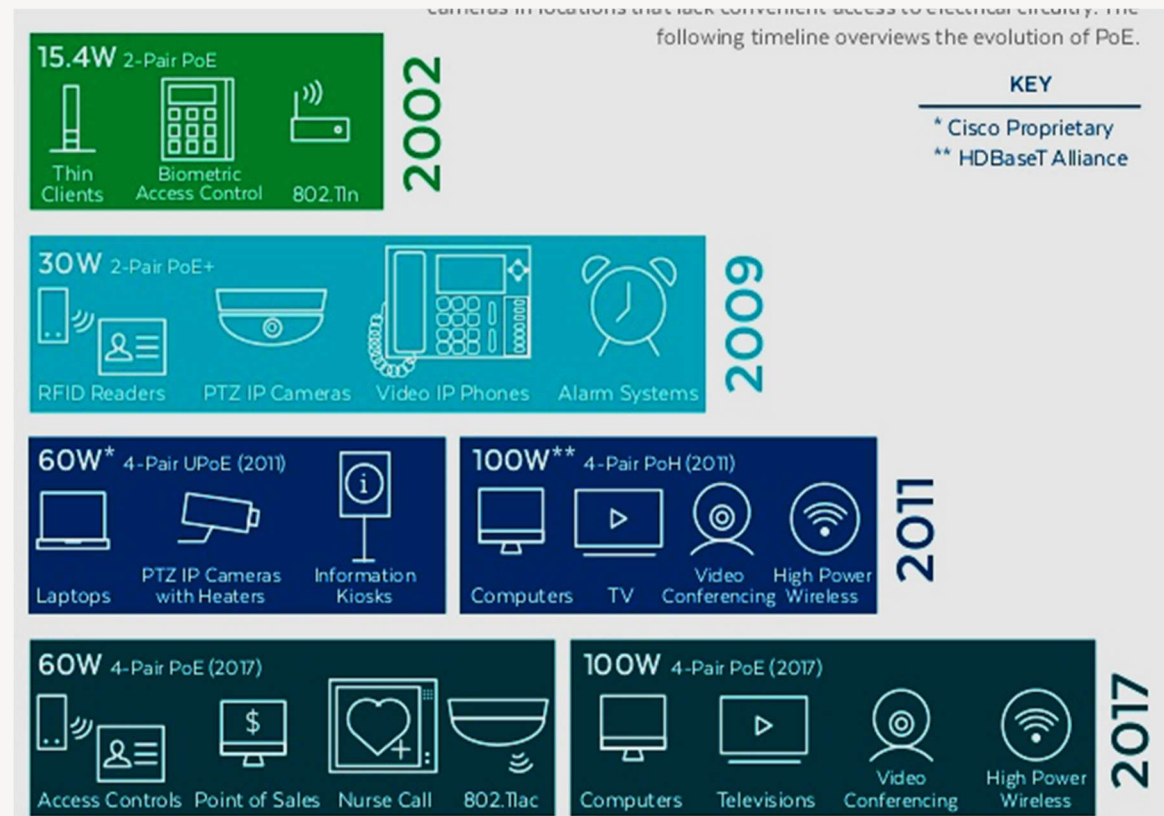


Name	Standard	Watts at PSE	Powered Wire Pairs
Cisco Inline Power	Cisco	7	2
PoE	802.3af	15	2
PoE+	802.3at	30	2
UPoE	802.3bt	60	4
UpoE+	802.3bt	100	4



Power Over Ethernet

- Mode A {pin 1,2,3,6}
- Mode B {pin 4,5,7,8}



TCP vs UDP



Function	Description
Multiplexing using ports	Function that enables receiving hosts to choose the correct application for which the data is destined, based on the destination port number.
Error recovery (reliability)	Process of numbering and acknowledging data with Sequence and Acknowledgment header fields.
Flow control using windowing	Process that involves a sliding window size that the two end devices dynamically agree upon at various points during the virtual connection. The window size, represented in bytes, is the maximum amount of data the source will send before receiving an acknowledgment from the destination.
Connection establishment and termination	Process used to initialize port numbers and Sequence and Acknowledgment fields.
Ordered data transfer and data segmentation	A continuous stream of bytes from an upper-layer process that is “segmented” for transmission and delivered to upper-layer processes at the receiving device, with the bytes in the same order.

TCP Port



Port Number	Protocol	Application
20	TCP	FTP data
21	TCP	FTP control
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	UDP,TCP	DNS
67, 68	UDP	DHCP
69	UDP	TFTP
80	TCP	HTTP (WWW)
110	TCP	POP3
161	UDP	SNMP
443	TCP	HTTPS (SSL)
16384–32767	UDP	RTP-based voice (VoIP) and video



TCP vs UDP headers

TCP Segment Header Format

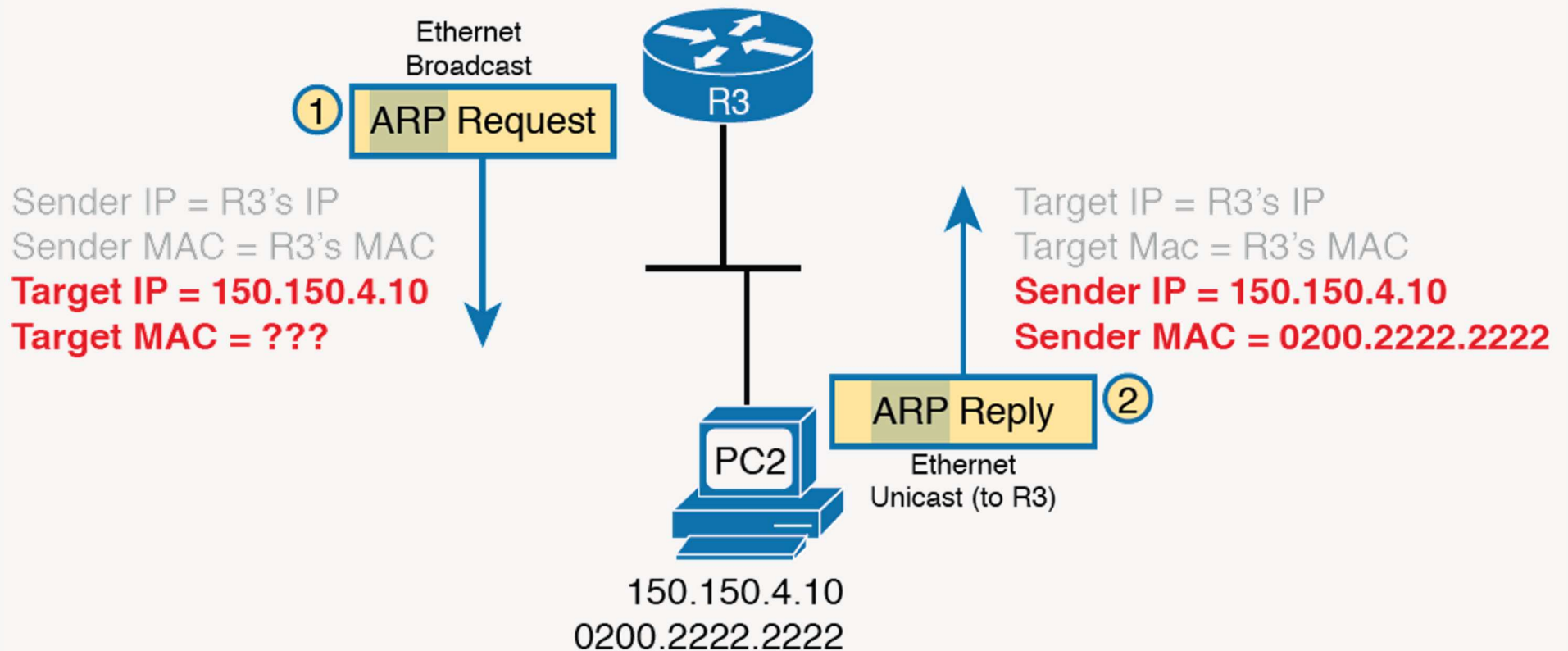
Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Sequence Number							
64	Acknowledgment Number							
96	Data Offset	Res	Flags		Window Size			
128	Header and Data Checksum				Urgent Pointer			
160...	Options							

UDP Datagram Header Format

Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Length				Header and Data Checksum			



ARP address resolution protocol



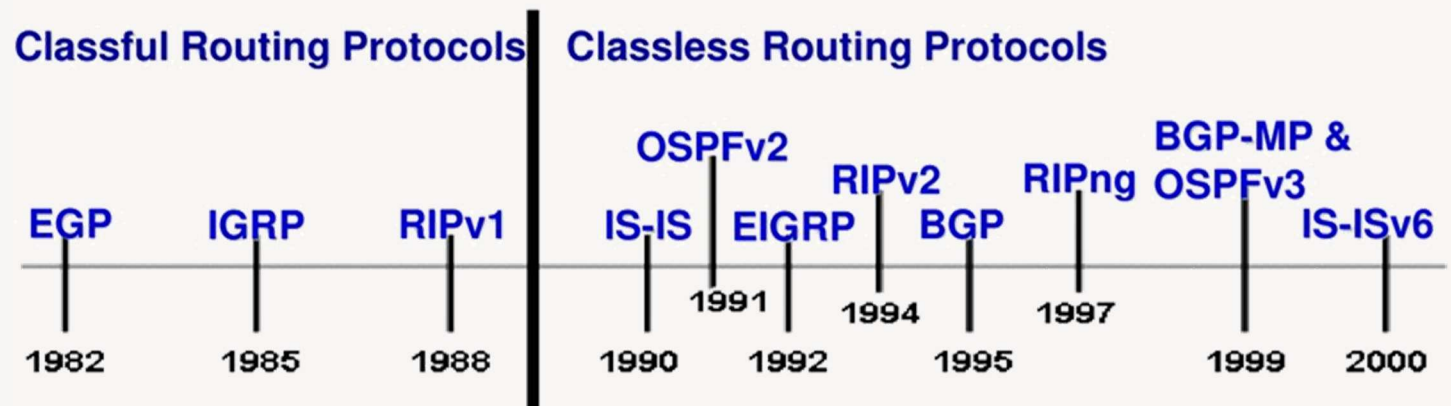
IP Address Management

- Subnetting (keep the network)
 - Supernetting (keep the host)
 - Reverse Engineering
 - VLSM
 - Route Summarization
-
- - Prevent waste of IP
 - - Better Management
 - - Faster Neighborhood / Forwarding
 - - Each VLAN use different subnets
 - - Router connects different subnets





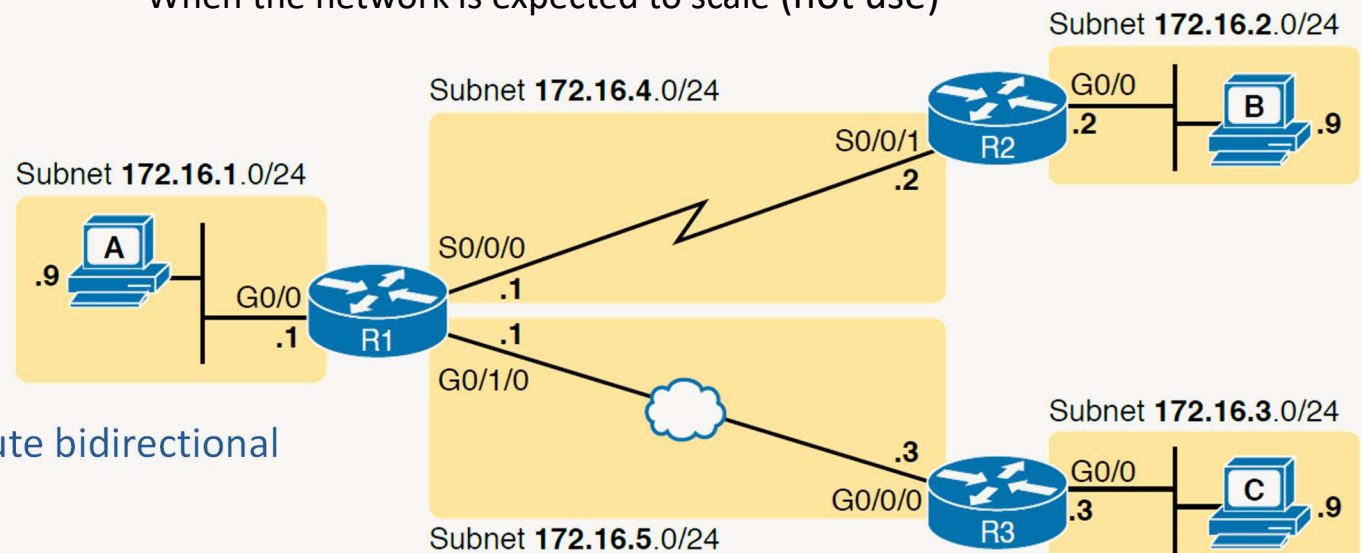
- Routing Protocols
 - Routed protocols
 - Autonomous System
 - IGP vs EGP
 - Static route and dynamic routes difference





Static and Default Route

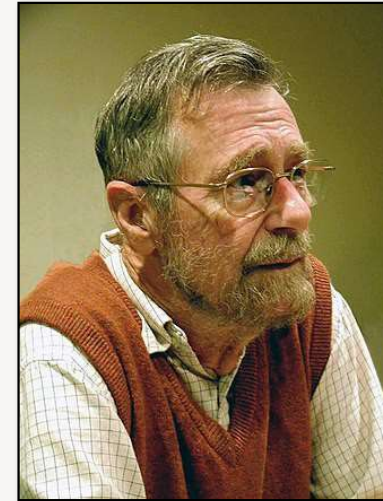
- Manual or static configuration
- For Small Networks
- Reliable connection
- Administrative Distance 1
- Mutual configuration
- Connected Network
- Non-connected Network
- Next Hop or exit interface
- Unidirectional config
- Stub Network
- To ISP site
- Gateway of last resort
- Do not configure Default route bidirectional
- In a small network that requires only simple routing
- In a hub-and-spoke network topology
- When you want to create a quick ad hoc route
- As a backup when the primary route fails
- In a large network (not use)
- When the network is expected to scale (not use)





Open Shortest Path First (OSPF)

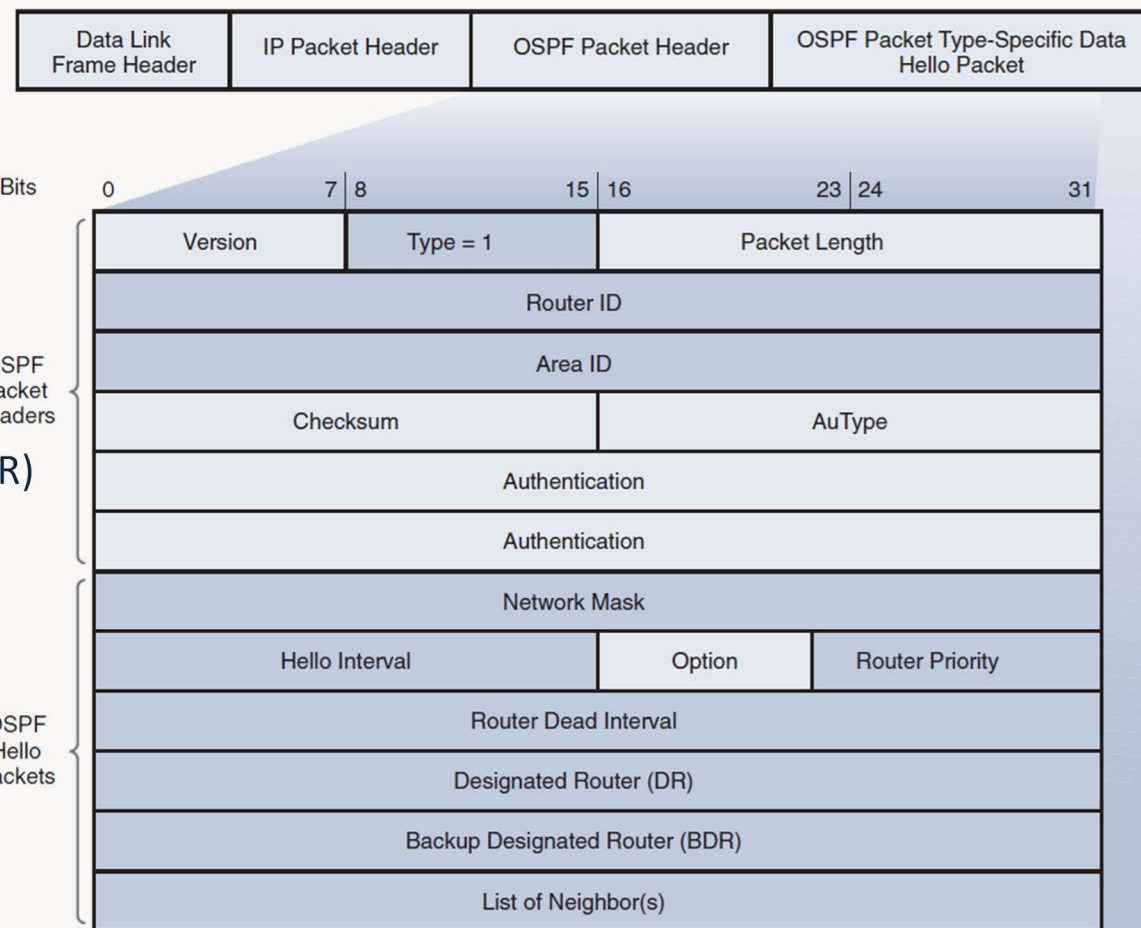
- IGP > Dynamic > Link-state
- Open Standard (IETF)
- Administrative Distance = 110
- 3 Table (Routing, Topology, Neighbor)
- Dijkstra Shortest Path First (SPF) algorithm
- Primary and Back up route
- Metric = cost
- Cost = $100/\text{Bandwidth}$
- Equal load balancing
- Default Equal Path = 4
- Hello: 10 Sec | Death: 40 Sec
- 30 sec 120 death on NBMA, FrameRelay, ATM networks



- Multicast : 224.0.0.5 & 224.0.0.6
- Multicast address: 01-00-5E-00-00-05 or 01-00-5E-00-00-06.
- IP packet header, the protocol field is set to 89 to indicate OSPF

OSPF cont....

- Area Base
- Area 0 Central Area
- Other Area should connect area 0
- Virtual Link
- Area Border Router (ABR)
- Autonomous System Boundary Router (ASBR)
- Manual Summarization on ABR and ASBR
- Designated Router and Backup DR
- OSPF Process-ID
- Wild Card Mask
- OSPF Router-ID
- Network cmd in OSPF





OSPF Cont....

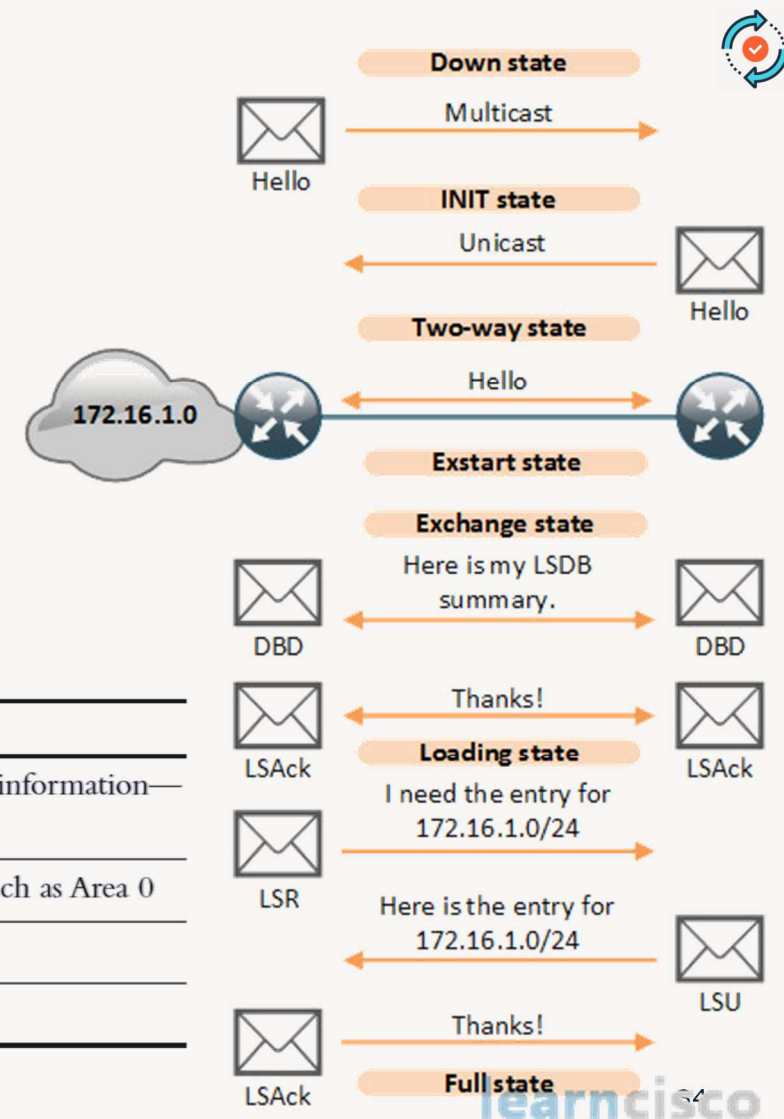
- Determine Router-ID (Highest physical, loopback int. or router-id)
- Add interface to Link State Database
- Send hello message on the interface
- Receive Hello packets
- Send Reply Hello (Rest timer or add as new neighbor)
- Master – Slave relationship
 - Master : the highest priority or router-id
 - Master sends DBD packets
 - Slave send its DBD
- DBD are acknowledge and received
 - Slave request detail (LSR)
 - Master send updates (LSU) & also vice versa.
- Neighbors are synchronized

OSPF Neighborship states

- Hello
- DBD: database description
- LSR: link state request
- LSU: link state update (contains LSAs)
- LSAck: link state acknowledge
- Hello (Type 1), DBD (Type 2), LS Request (Type 3), LS Update (Type 4), LS ACK (Type 5)

Term	Description
Area	A set of routers and links that shares the same detailed LSDB information—but not with routers in other areas—for better efficiency
Backbone area	A special OSPF area to which all other areas must connect, such as Area 0
Intra-area route	A route to a subnet inside the same area as the router
Interarea route	A route to a subnet in an area the router is not a part of

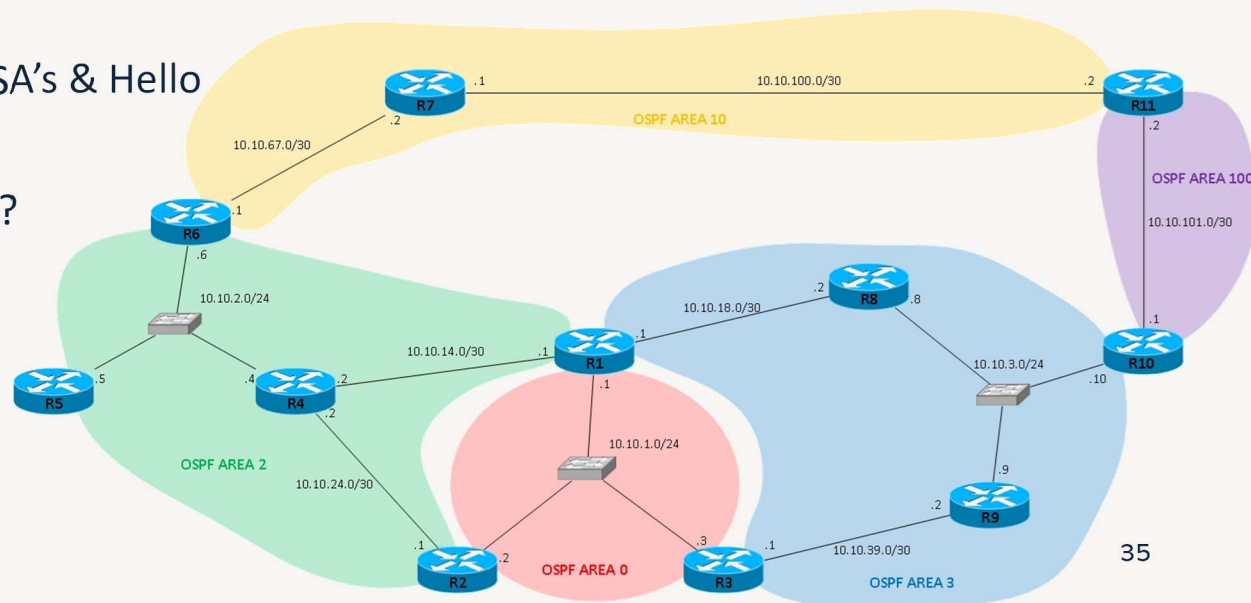
Copyright © ZoomByte | CCNA





OSPF Advanced

- OSPF neighbors refresh routing protocols every 30 min
- OSPF use Link State Advertisement for topology changes
- OSPF LSA contain: Prefix/metric/router-id/source ...
- All internal routers must have the same image of the network;
- (SPF) is run on link state database (LSDB) to find best, lowest cost, paths to all destinations.
- Auto-cost reference bandwidth
- Build & maintain OSPF database by LSA's & Hello
- Contiguous network
- Two OSPF neighbors must have same?
- Wireshark capture hello packets
- Passive interface
- Default-information originate





OSPF Neighborship types

- OSPF has two neighborship classes:
- 2-Way Neighbors
- Fully adjacent neighbors
- Hello & death timer modifying

Type	Packet Name	Description
1	Hello	Discovers neighbors and builds adjacencies between them.
2	DBD	Checks for database synchronization between routers.
3	LSR	Requests specific link-state records from router to router.
4	LSU	Sends specifically requested link-state records.
5	LSAck	Acknowledges the other packet types.



The acronyms LSA and LSU are often used interchangeably.

An LSU contains one or more LSAs.

LSAs contain route information for destination networks.

LSA specifics are discussed in CCNP.

LSA Type	Description
1	Router LSAs
2	Network LSAs
3 or 4	Summary LSAs
5	Autonomous System External LSAs
6	Multicast OSPF LSAs
7	Defined for Not-So-Stubby Areas
8	External Attributes LSA for Border Gateway Protocol (BGP)
9, 10, 11	Opaque LSAs



OSPFv2 and OSPFv3 Difference

Feature	OSPFv2	OSPFv3
Advertising	IPv4 networks	IPv6 prefixes
Source address	IPv4 source address	IPv6 link-local address
Destination address	Choice of Neighbor IPv4 unicast address 224.0.0.5, all-OSPF-routers multicast address 224.0.0.6, DR/BDR multicast address	Choice of Neighbor IPv6 link-local address FF02::5, all-OSPFv3-routers multicast address FF02::6, DR/BDR multicast address
Advertising networks	Configured using the network router configuration command	Configured using the ipv6 ospf area interface configuration command
IP unicast routing	IPv4 unicast routing enabled by default	Requires configuration of the ipv6 unicast- routing global configuration command
Authentication	Plain text and MD5	IPsec



OSPF network types

- Router(config-if)# **ip ospf priority {0 - 255}**
- R1# **clear ip ospf process**
- Router# **Show ip ospf interface**
- Router(config-subif)# **ip ospf network broadcast**

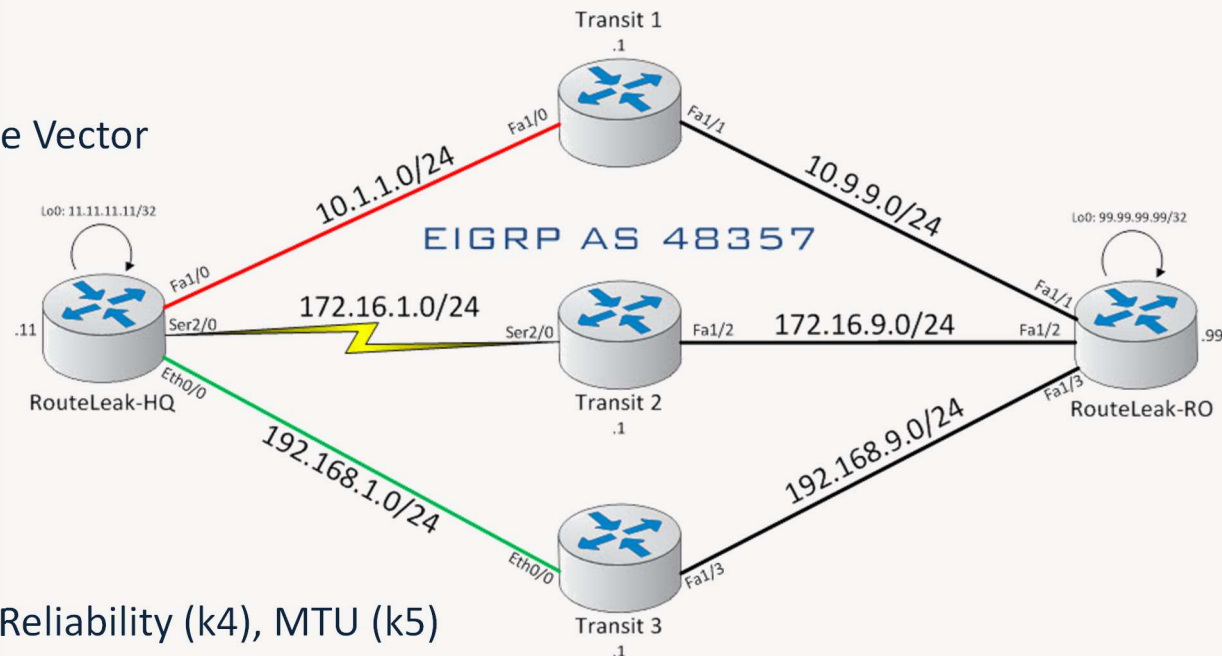
Network Type	Elects a DR and BDR	Default Hello Interval	Uses the neighbor Command
Broadcast	Yes	10 Seconds	No
Point-to-Point	No	10 Seconds	No
Non-Broadcast(NBMA)	Yes	30 Seconds	Yes
Point-to-Multipoint	No	30 Seconds	No



Enhanced Interior Gateway Routing Protocol (EIGRP)



- Cisco Routing Protocol
- Open standard since March 2013, IETF since 2016
- Driven From IGRP
- Support IPv4, IPv6, AppleTalk, IPx
- IGP > Dynamic > Hybrid or Adv. Distance Vector
- 3 Table (Routing, Topology, Neighbor)
- Interface or connected network table
- keep backup route
- Successor and Feasible successor
- DUAL= Diffusing Update Algorithm
- Metric = K calculation
- **Bandwidth (k1)**, Load (k2), **Delay (k3)**, Reliability (k4), MTU (k5)





EIGRP cont...

- Unequal Load balancing
- Hello = 5 / Hold Down = 15 | 60/180 on T1 connection
- Multicast: 224.0.0.10 – MAC : 0100.5E00.000A – IPv6 : FF02::A
- Neighbors interface primary IP must be same.
- Autonomous system base (must be same)
- Auto-summary is enable by default
- Maximum Hop count = 255 and default= 100
- Route Summarization base on interface
- Support: Multicast, VLSM, Authentication (MD5), Classless
- Two EIGRP routers: Same AS# | K value | Subnet | authentication
- Fast convergence
- AD: internal: 90 | external: 170
- Full update for new neighbor, partial update for new changes
- Dynamically discover other EIGRP routers
- Reliable Transfer Protocol (RTP) – IP header 88
- EIGRP timers can be different at two routers
- Configuration Review:
 - 1) neighboring
 - 2) discover neighbor by multicast
 - 3) advertise networks
 - 4) wildcard mask or no-auto summary



Routing Protocols parameters

3.1 Interpret the components of routing table

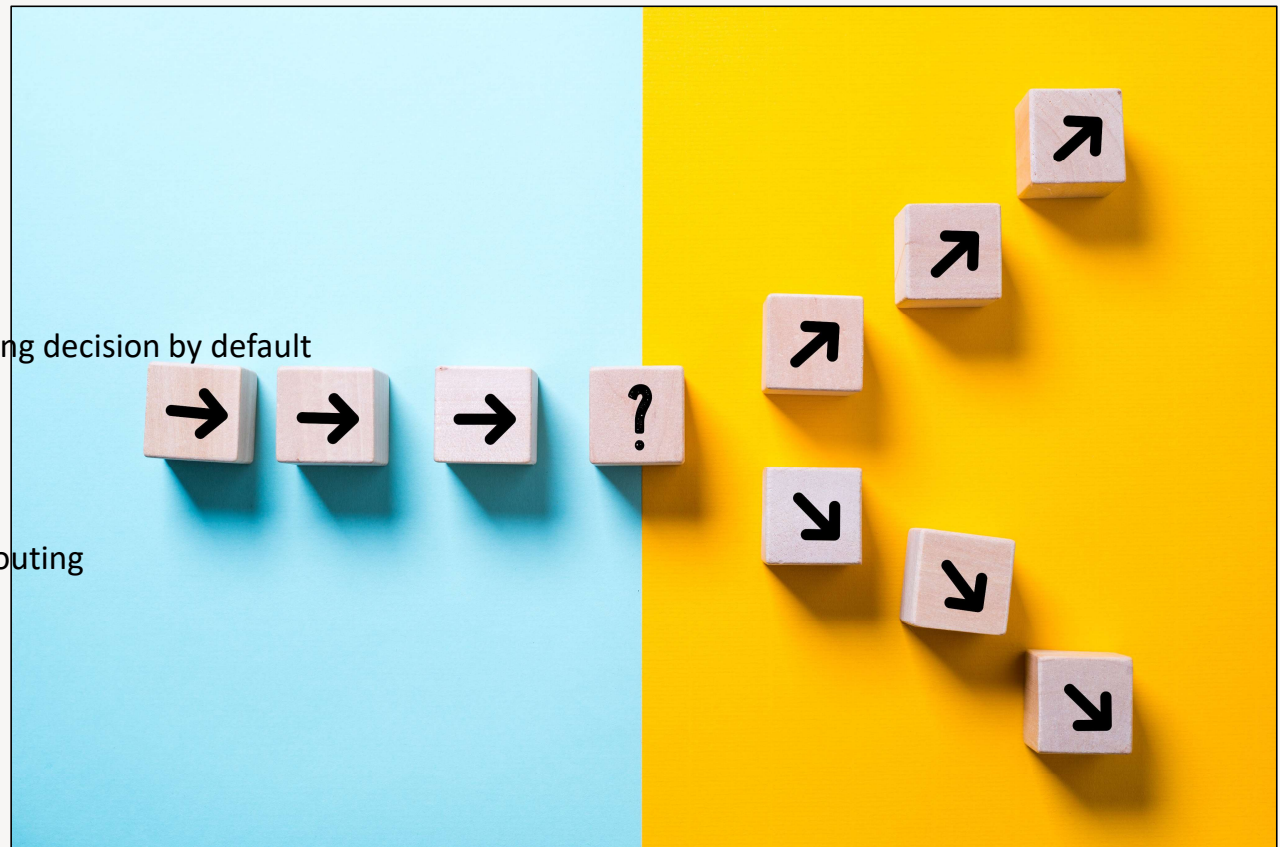
- 3.1.a Routing protocol code
- 3.1.b Prefix
- 3.1.c Network mask
- 3.1.d Next hop
- 3.1.e Administrative distance
- 3.1.f Metric
- 3.1.g Gateway of last resort

3.2 Determine how a router makes a forwarding decision by default

- 3.2.a Longest match
- 3.2.b Administrative distance
- 3.2.c Routing protocol metric

3.3 Configure and verify IPv4 and IPv6 static routing

- 3.3.a Default route
- 3.3.b Network route
- 3.3.c Host route
- 3.3.d Floating static



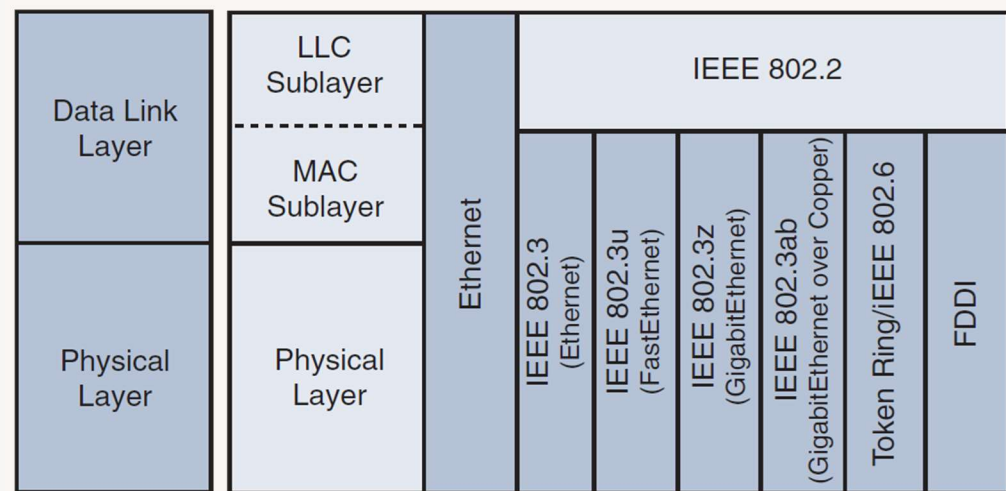


Switching Concept

- Symmetric and Asymmetric Switching
- Memory Buffering (Port-based memory | Shared memory)
- Logical Link Control (LLC) sublayer: Defined in the 802.2 standard
- Media Access Control (MAC) sublayer: Defined in the 802.3 standard
- CSMA/CD | CSMA/CA | Collision & Broadcast domain
- Frame switching
- MAC address table

Switching forwarding methods:

- ✓ Store-and-forward switching
- ✓ Cut-through switching
- ✓ Fragment-free mode (64 Bytes)



OSI Layers

LAN Specification

Figure 30-5 TIA/EIA Standard Ethernet Cabling Pinouts

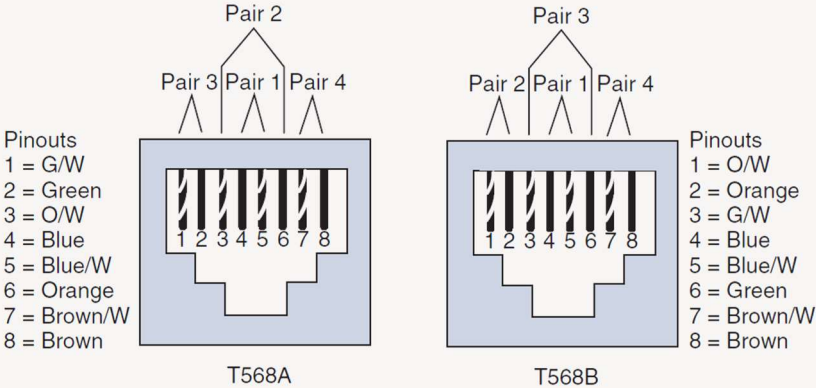


Table 30-2 10BASE-T and 100BASE-TX Pin Pairs Used

Devices That Transmit on 1,2 and Receive on 3,6	Devices That Transmit on 3,6 and Receive on 1,2
PC NICs	Hubs
Routers	Switches
Wireless access points (Ethernet interfaces)	—
Networked printers (printers that connect directly to the LAN)	—

Table 30-1 Today's Most Common Types of Ethernet

Common Name	Speed	Alternative Name	Name of IEEE Standard	Cable Type, Maximum Length
Ethernet	10 Mbps	10BASE-T	802.3	Copper, 100 m
Fast Ethernet	100 Mbps	100BASE-TX	802.3u	Copper, 100 m
Gigabit Ethernet	1000 Mbps	1000BASE-LX	802.3z	Fiber, 550 m
Gigabit Ethernet	1000 Mbps	1000BASE-T	802.3ab	Copper, 100 m
10GigE (Gigabit Ethernet)	10 Gbps	10GBASE-T	802.3an	Copper, 100 m
10GigE (Gigabit Ethernet)	10 Gbps	10GBASE-S	802.3ae	Fiber, 400 m



Ethernet Framing

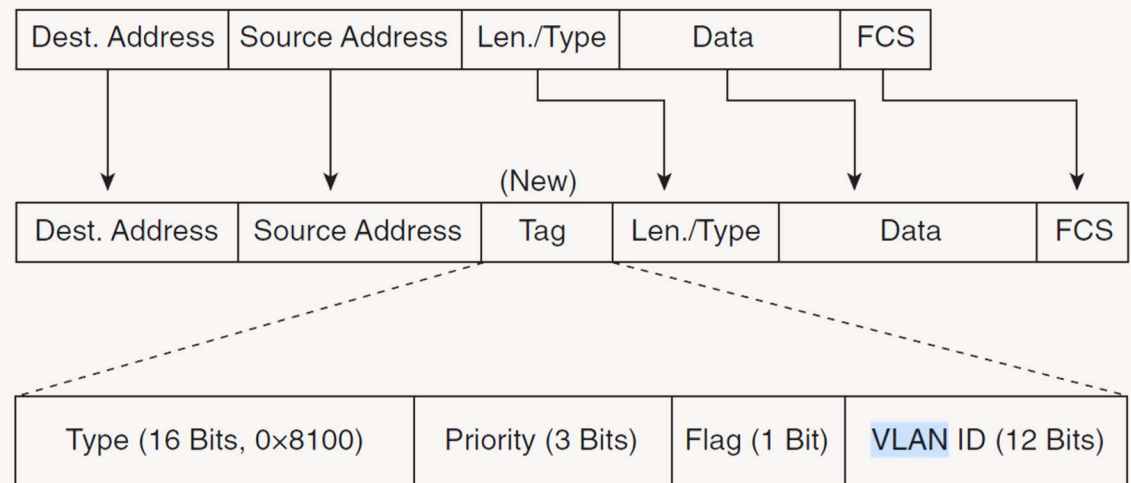
Field	Field Length, in Bytes	Description
Preamble	7	Synchronization
Start Frame Delimiter (SFD)	1	Signifies that the next byte begins the Destination MAC field
Destination MAC Address	6	Identifies the intended recipient of this frame
Source MAC Address	6	Identifies the sender of this frame
Length	2	Defines the length of the data field of the frame (either length or type is present, but not both)
Type	2	Defines the type of protocol listed inside the frame (either length or type is present, but not both)
Data and Pad	46–1500	Holds data from a higher layer, typically a Layer 3 PDU (generic) and often an IP packet
Frame Check Sequence (FCS)	4	Provides a method for the receiving NIC to determine whether the frame experienced transmission errors



Virtual Local Area Network

- Switching Feature
- Logically separate our LAN
- Small Broadcast domain
- Better QoS, Security and Management
- Cost reduction
- Each VLAN require different subnet
- VLAN 1 (Default)
- VLAN Number Range: 0 4095
- Primary VLAN Range: 1 1005
- Extended VLAN Range: 1006 4094
- Reserved VLAN Numbers: 0, 4095 , 1002, 1003, 1004, 1005
- VLAN Database { Flash:vlan.dat}

Figure 26-3 Fields of the 802.1Q Tag Inside an Ethernet Frame





VLAN Types & DTP

- Data VLAN
- Voice VLAN
- Management VLAN
- Black Hole VLAN / Death VLAN
- Native VLAN
- Default VLAN (1 , 1002, 1003, 1004, 1005)
- Wireless VLAN
- Cisco protocol
- Negotiation for trunking
- Auto and Desirable mode
- VLAN filtering on trunk

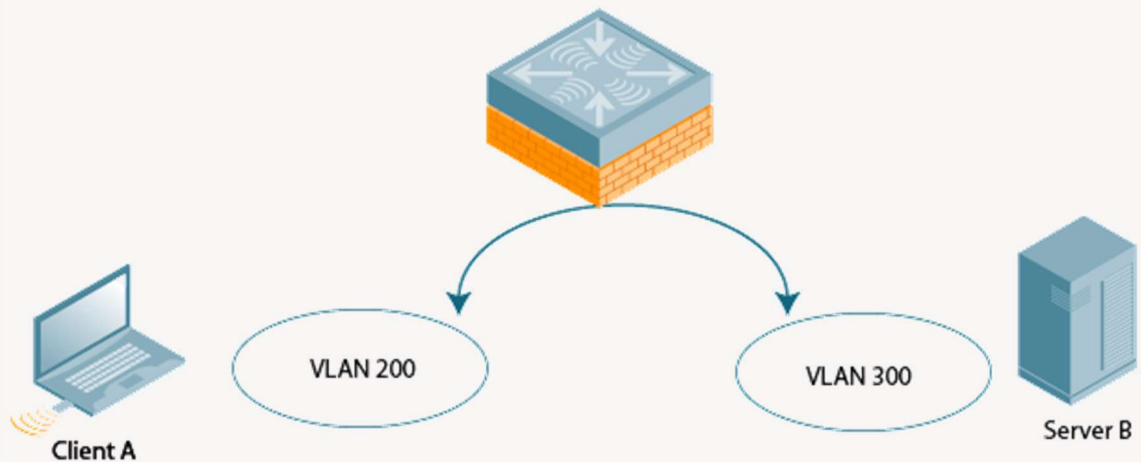
VLAN connectivity

- Switchport modes:
 - Access: end user – one VLAN data transaction
 - Trunk : connectivity devices – switch , WAP, Router – multi VLAN data transaction
 - Dynamic : auto – Access – Access – Trunk Trunk



Routing Different VLANs

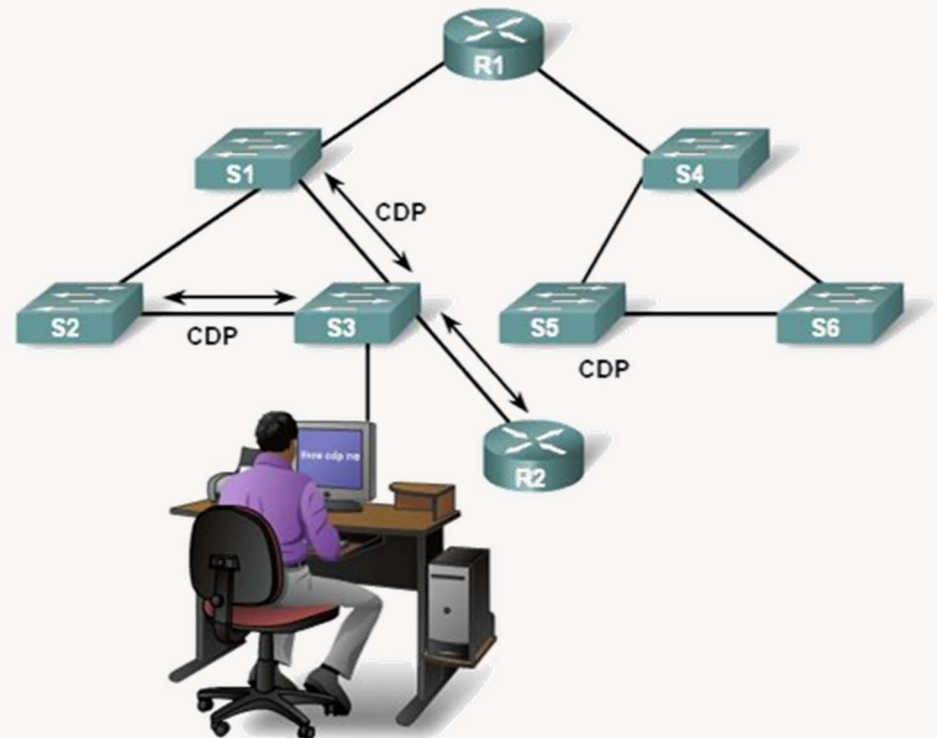
- Inter VLAN Routing
- Inter VLAN Routing (Load Balancing mode)
- Router on A Stick
- Switch Virtual Interface (SVI)





Cisco Discovery Protocol

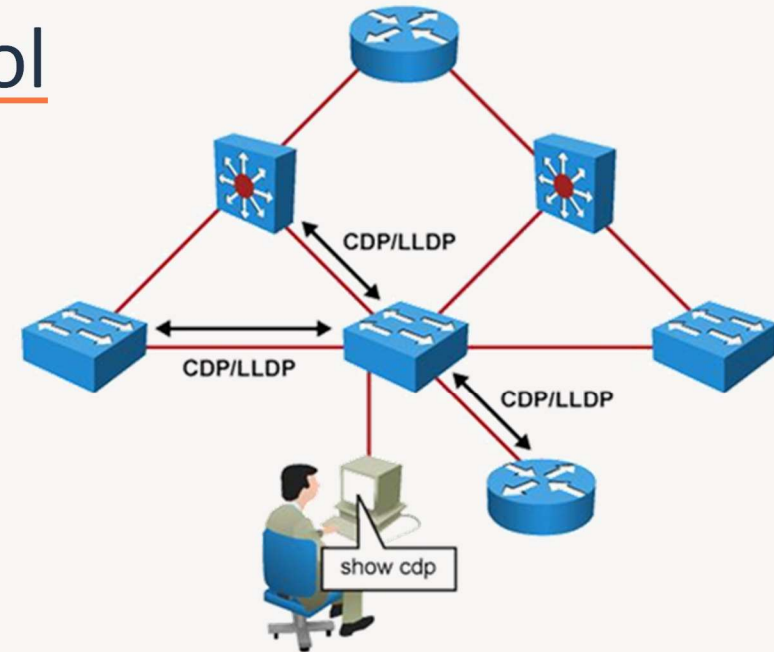
- Cisco proprietary protocol
- Identify/Determine the cisco directly connected
- Works at Data-Link layer (Layer2)
- No need for IP address just ports should be up
- 60 seconds interval, hold timer 180 seconds
- Shows the neighbor Device ID – Address – Port ID – Capabilities – Version - Platform IOS, Device Type, Duplex, VLANs.
- Enabling and disabling CDP globally and interface.
- While the CDP Version 1 prohibits native VLAN information to pass between Cisco switches, Version 2 can pass native VLAN information.
- `cdp timer seconds`
- `cdp holdtime seconds`





Link Layer Discovery Protocol

- Similar to CDP protocol
- IEEE 802.1ab open standard protocol
- Type-Length Value (TLV)
- 30 Sec Interval
- 120 Hold Down
- `lldp timer seconds`
- `lldp holdtime seconds`



Parameter	Default
LLDP	Disabled globally and on all interfaces
LLDP timer	30 seconds
LLDP holdtime	120 seconds
LLDP reinitialization delay	2 seconds



Spanning Tree Protocol

- Ethernet Bridging loop
- Switching Redundancy Feature
- IEEE 802.1D open standard
- Single Point of Failure
- Hierarchical design
- Root Bridge
- Bridge ID (2 Byte Priority + 6 Byte MAC)
- BPDU Bridge Protocol Data Unit
- STP ports mode (Designated, Root, Block port)
- STP ports Status (Disable > Listening > Learning > Forwarding / Blocking)
- Root Election
- STP port cost
- **Broadcast storms:** Each switch floods broadcasts endlessly.
- **Multiple-frame transmission:** Multiple copies of unicast frames are delivered to the destination, causing unrecoverable errors.
- **MAC database instability:** Instability in the content of the MAC address table results from different ports of the switch receiving copies of the same frame.

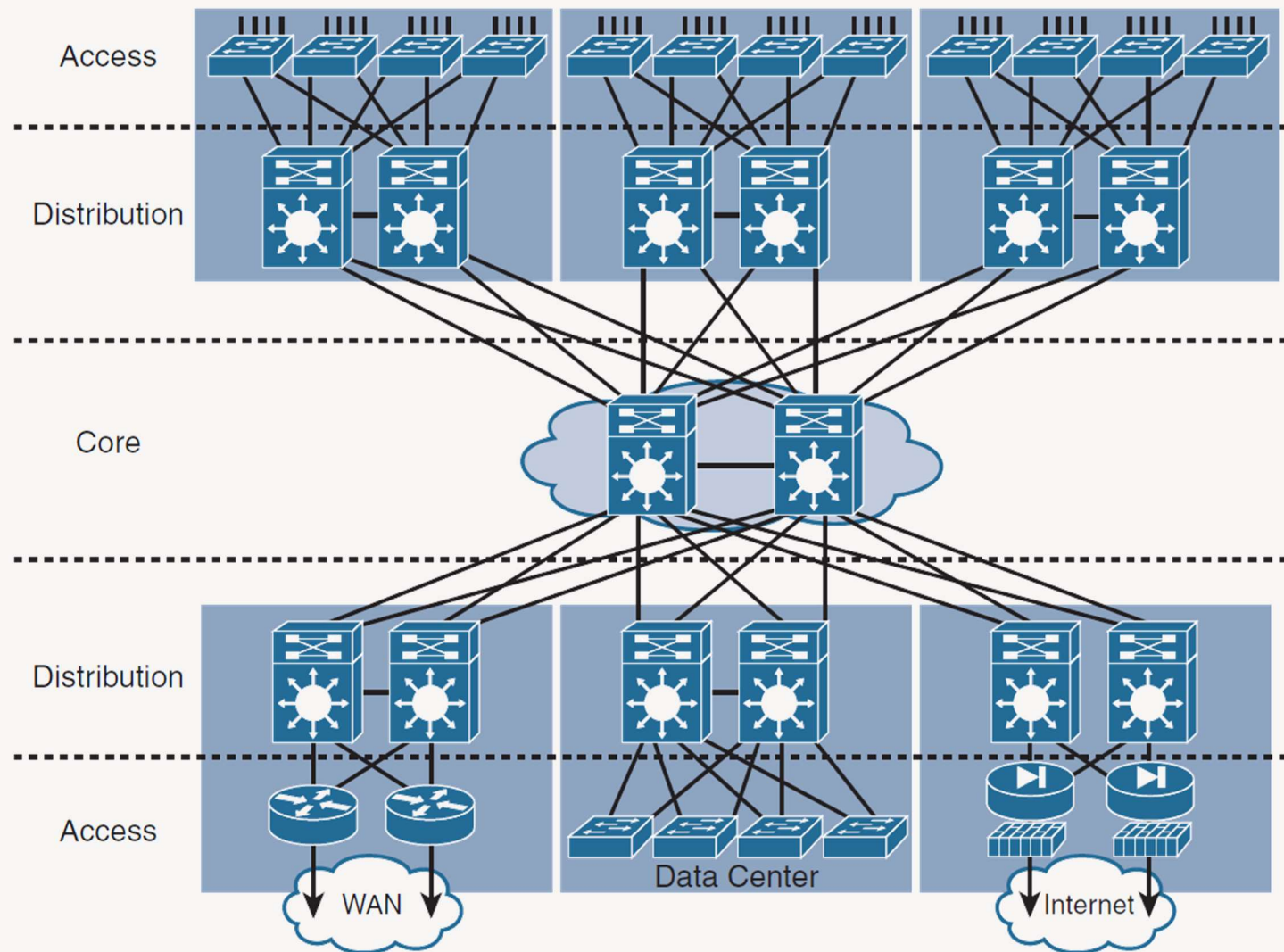


STP advanced concepts

- Only Root BPDU every 2 seconds
- BPDU multicast: 01-80-c2-00-00-00
- Two type of BPDU: configuration BPDU, TCN BPDU
- STP root port election (Lowest port cost, bridge ID, Port#, port priority)
- STP Timers (BPDU: 2 Sec, Forwarding Delay: 15 Sec, Max Age: 20)
- Seven Switches Drives these values
- STP port priority 0-255 (128)
- Modifying STP Timers

Ethernet Speed	Original IEEE Cost	Revised IEEE Cost
10 Mbps	100	100
100 Mbps	10	19
1 Gbps	1	4
10 Gbps	1	2

Switch Hierarchical design



STP Types & solutions



IEEE → STP → 802.1D → loop prevention → 30 — 50 sec

Cisco → Port fast → No list/Leav → forwarding → End user

Cisco → BPDUGuard → error Disable → shutdown

Cisco → BPDUFILTER → Not Down → Filter BPDU → MAC flapping

Cisco → Error Disable Recovery → 15 - 30 - 55 sec

Cisco → uplink fast — Back Bone Fast — 50 — 10 sec

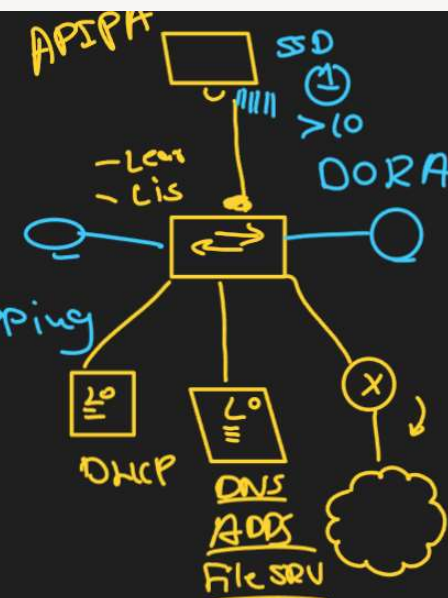
Cisco → Per VLAN STP → ISL — Trunk — 30 Byte

Cisco → Per VLAN STP + → ISL — 802.1Q

IEEE → Rapid STP + Port fast → > 5 sec

* Cisco → Rapid Per VLAN STP + > 5

IEEE → MSTP → For more VLANs



IEEE → Wifi 6
14 Gb/s

Cisco



PortFast & BPDU Guard

- Disable listening and learning
- Ports directly connected to the end users and host should be portfast
- Port with portfast will directly going to forwarding state
- Edge port on RSTP
- BPDU Guard concept & Config

Spanning-tree portfast default {first do the trunking}

Spanning-tree portfast

Switchport host (Access,Portfast,BPDUGuard)





Rapid STP

- IEEE open standard 802.1w
- Design to speed up convergence
- Bi-direction BPDUs sending
- BPDUs are now use a keepalive message
- Hello intervals 2 max age 6 seconds but 802.1D 20 max age
- Full duplex link is consider P2P
- Half duplex link is consider shared
- Root port, Designated port,
- Backup port (Hub)
- Alternative port (switch)
- Discarding (Disabled, blocking and listening)



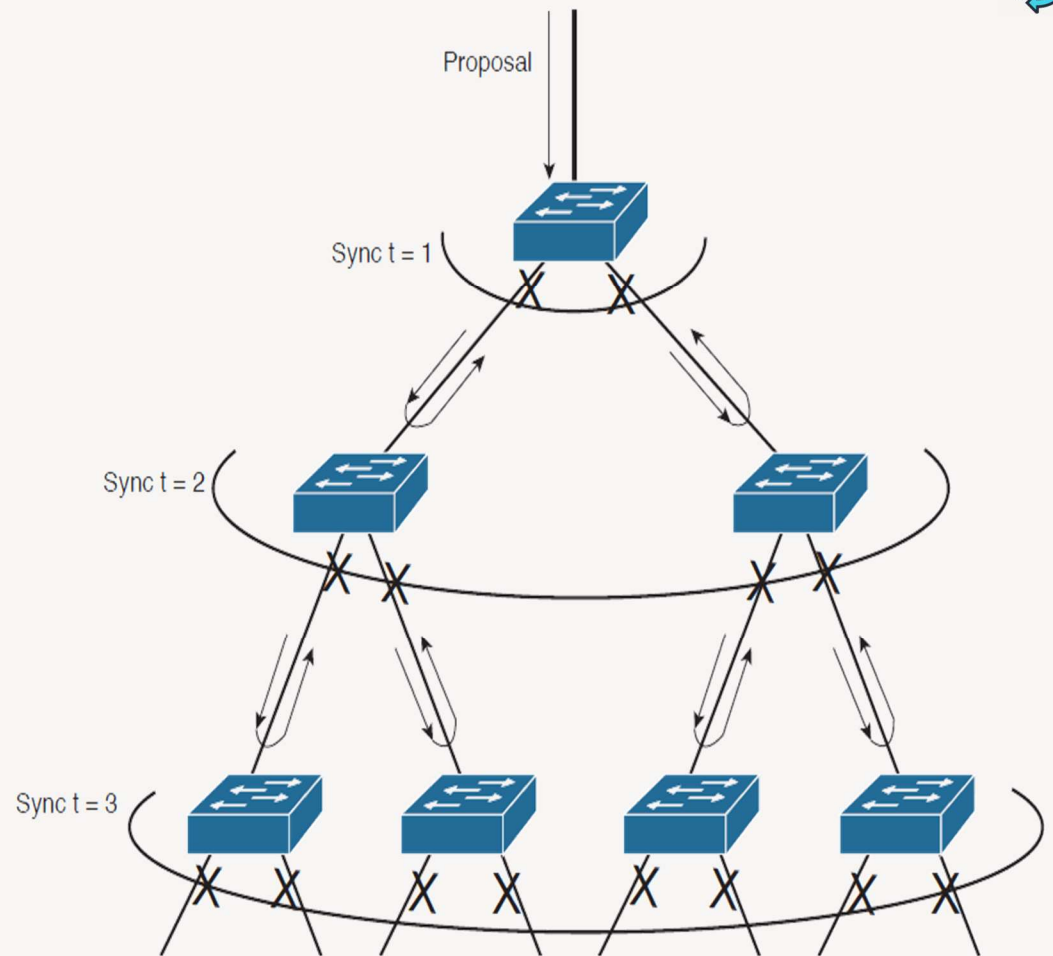
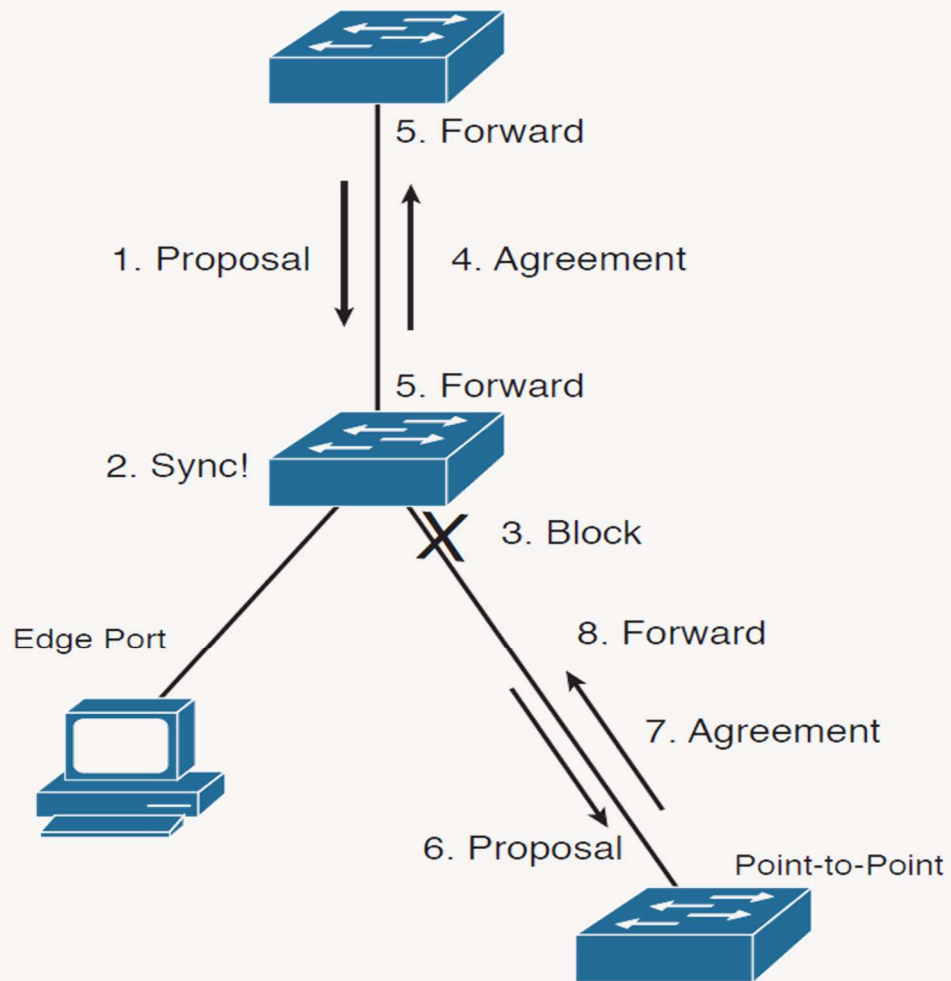


Rapid Spanning Tree BPDUs



- Learning and forwarding still here
- Built-in uplink fast, Edge port (Portfast) and back-bonefast
- BPDU Flags (proposal and agreement) non-edge ports
- Don't use time, base on negotiation

STP	RSTP
Disabled	Discarding
Blocking	
Listenning	
Learning	Learning
Forwarding	Forwarding





EtherChannel

- Redundancy Feature
- Aggregate multiple physical connection
- Fast Performance , No Ethernet loop
- Logically one interface , Port-channel
- Up to 2-8 connection
- All ports must have the same: speed , duplex, access, trunk, protocol
- Load balancing : per connection / stream
- Two protocol: Link aggregation control protocol (IEEE) 802.3ad
- Port Aggregation protocol (Cisco)
- Port-channel configuration mostly not ports
- Implementation Restrictions in 2960 series
- Cannot be mixed Fast & Gig within the same
- Each EtherChannel can consist of up to eight compatibly ports.
- Cisco IOS Software currently supports up to six EtherChannels.
- The EtherChannel configuration must be consistent on the two switches. The trunking configuration (native VLAN, allowed VLANs, and so on) must be the same. All ports also must be Layer 2 ports.
- All ports in the EtherChannel must be Layer 2 ports, or all ports within the EtherChannel must be Layer 3 ports.



L3 EtherChannel & Misconfiguration Guard

- Etherchannel port share single MAC address & port-id
- Enable by default
- Detects by port-id whether it is multiple port-id no etherchannel
- Place ports in err-disable

Port-channel load-balance ?

Spanning-tree etherchannel guard misconfig

Show spanning-tree summary

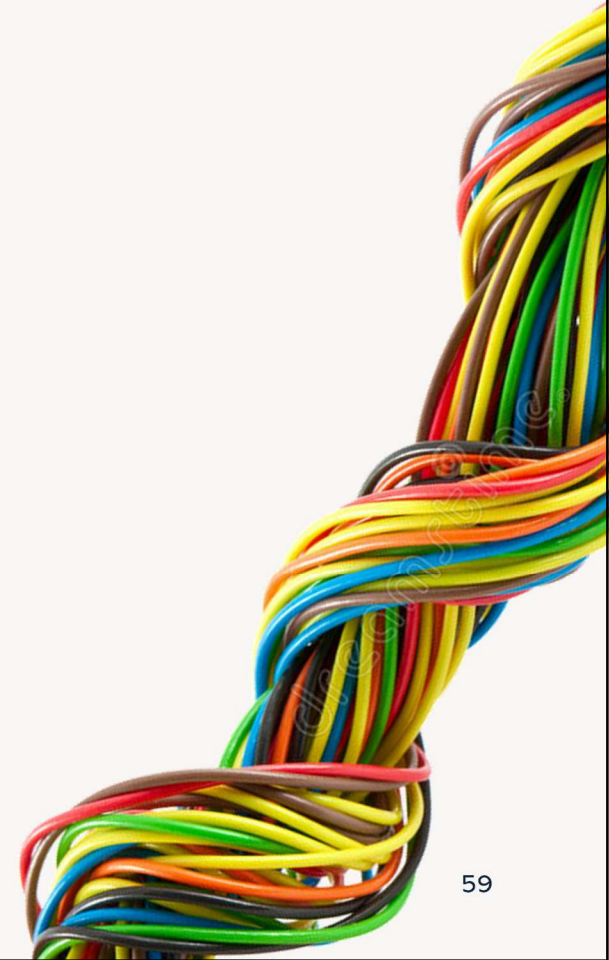




Table 24-2 LACP Mode Settings

Sw1	Sw2	Channel Established?
On	On	Yes
Active/Passive	Active	Yes
On/Active/Passive	Not configured	No
On	Active	No
Passive/On	Passive	No

Match Settings	
Port type	Duplex
Port mode	MTU
Native VLAN	Load interval
Allowed VLAN	Storm control
Speed	

Table 24-1 PAgP Mode Settings

Sw1	Sw2	Channel Established?
On	On	Yes
Auto/Desirable	Desirable	Yes
On/Auto/Desirable	Not configured	No
On	Desirable	No
Auto/On	Auto	No